Updated Model for Bonded Anchor Concrete Edge Failure Tests

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1 Introduction

Regarding the concrete edgefailure tests for bonded anchors the model in Figure 1 is given for the process of virtual approval. In the Figures and throughout the text quantities with indices $(\cdot)_{\mathrm{c}}, (\cdot)_{\mathrm{s}}, (\cdot)_{\mathrm{m}}, (\cdot)_{\mathrm{a}}$ indicate geometric/ material properties of the concrete, steel, mortar and anchor respectively.

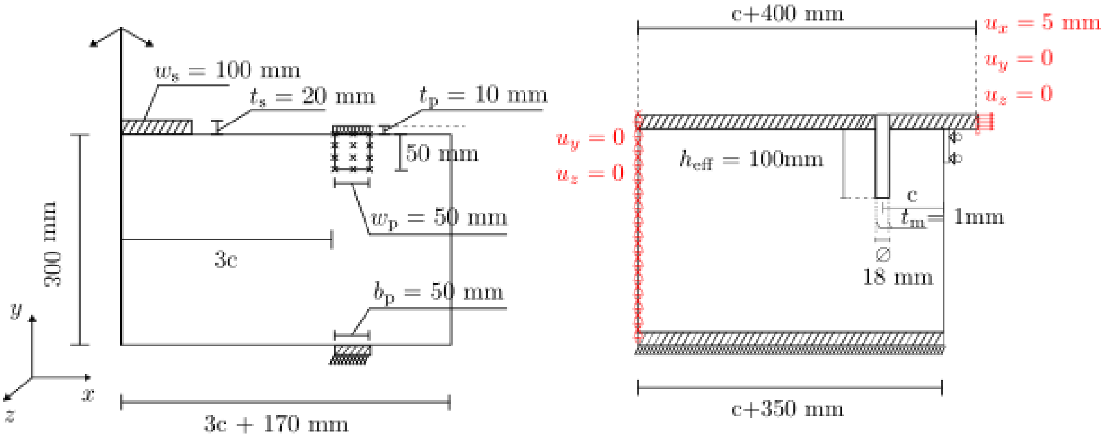

Figure 1: Geometry of the bonded anchor model for concrete edge failure. Left: front view of the model including the symmetry plane. Right: Section view of the model.

1.1 Geometry

The model dimensions depend on the distance of the anchor from the concrete edge c, which we treat here as a parameter. Moreover, the model is symmetric along the z plane, thus we describe only half of it. The model consists of a cylindrical steel anchor with diameter $d_{\mathrm{a}} = 16$ mm, embedded to a depth of $h_{\mathrm{eff}} = 100$ mm, in an orthogonal parallelepiped concrete slab with dimensions (length: $L_{\mathrm{c}} =$ c+350 mm, width: $W_{\mathrm{c}} =$ 3c+170 mm, height: $H_{\mathrm{c}}=$ 300 mm). A cylindrical hole is present inside the concrete slab with diameter $d_{\mathrm{h}} = d_{\mathrm{a}} + 2t_{\mathrm{m}} = 18$ mm and $t_{\mathrm{m}} = 1$ mm is the mortar thickness. No modeling of the drilling process or any of the residual stresses that may develop takes place. The anchor is placed inside the hole so that their axes along the height match. The mortar region is then modeled as a hollow cylinder sleeve with height h_{eff} and constant thickness t_{m} between the anchor and the surrounding concrete.

The anchor is connected to a steel plate on top with dimensions (length: $b_{\mathrm{s}} = c + 400$ mm, width: $w_{\mathrm{s}} = 100$ mm, height: $t_{\mathrm{s}} = 20$ mm). The plate contains a hole in the middle with diameter matching the diameter of the anchor, through its thickness. The connection of the anchor to the plate is as follows: First, the anchor is extended above the concrete slab by a length equal to the thickness of the steel plate. Second, the anchor extension is placed in the center of the steel plate hole. The two components (anchor, plate) then are merged together behaving as one body for subsequent discretizations of the model.

The assumed loading and the model are symmetric regarding the global z coordinate thus only half of the model needs to be simulated (see also Figure 2).

the component dimensions are summarized in the next table:

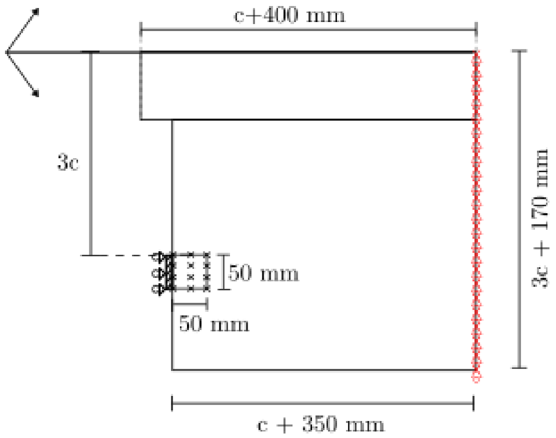

Figure 2: Top view of the model including the plane of symmetry across which the model and the loading are symmetric.

Parameter	Value	Description
c	50|| 100 || 150	edge distance
d_{a}	16 mm	Anchor diameter
h_{eff}	100 mm	Effective embedment depth
t_{m}	1 mm	Mortar thickness
L_{c}	c+350 mm	Slab length
W_{c}	3c+170 mm	Slab width of the half-model
H_{c}	300 mm	Slab height
b_{s}	400 + c mm	Plate length
w_{s}	100 mm	Plate width
t_{s}	20 mm	Plate thickness

Table 1: Geometrical parameters of the bonded anchor model. Three values of the concrete edge distance are to be taken into account.

1.2 Material properties

The concrete properties are given in Table 2, i.e grade C20/25. In the Table: E_{c} denotes the Young's modulus, ν_{c} denotes the Poisson's ratio, f_{cy} denotes the yield strength, f_{cu} denotes the ultimate strength, f_{bu} denotes the concrete ultimate biaxial strength, f_{tu} denotes the concrete ultimate tensile strength, $G^{\mathrm{f}}_{\mathrm{Ic}}$ is the concrete fracture energy in mode one fracture, and ρ_{c} is the density of concrete.

The anchor and steel plate are made from structural steel with known elastic properties and density see Table 3. The anchor is to be modeled elastoplastic with a yield strength $f_{\mathrm{ya}} = 1080$ MPa. In the Table: E_{s} denotes the Young's modulus, ν_{s} denotes the Poisson's ratio and ρ_{s} is the density of steel.

Mortar properties are given in Table 3. For edge failure tests failure in mortar is not critical therefore assume mortar to be elastic. Mortar stiffness differs in the normal and tangential directions. $E_{\mathrm{nm}}, G_{\mathrm{nt}}$.

1.3 Boundary conditions

A uniform horizontal displacement boundary condition ($u_x = 5$ mm) is applied over the total surface of the right part of the steel plate (see also right part of Figure 1). The vertical and through thickness displacements u_y, u_z are set to zero in order to model the hydraulic jack.

The back part of the steel plate is constrained in the vertical and through thickness directions to represent the clamps.

The back part of the concrete slab is constrained in the vertical and through thickness directions to

Parameter	Value	Units
E_c	3.06×10^4	MPa
ν_c	0.2	-
f_cy	12.9	MPa
f_cu	32.3	MPa
f_bu	37.5	MPa
f_tu	2.72	MPa
G^f_Ic	95.3×10^{-3}	N/mm
ρ_c	2.4×10^{-6}	kg/mm^3

Table 2: Concrete properties, estimated from [3]

Parameter	Value	Units
E_s	2×10^5	MPa
ν_s	0.33	-
ρ_s	7.85×10^{-6}	kg/mm^3
f_ya	1080	MPa
E_m	5400	MPa
ρ_m	2×10^{-6}	kg/mm^3

Table 3: Steel and mortar properties

represent the continuity of the slab.

The bottom part of the concrete slab is connected through frictionless contact (no penetration is allowed) to a steel plate running along the length of the concrete slab, in order to model the connection of the slab to the timber beams in the original experiment. The bottom steel plate is fixed to the ground.

In the front part of the concrete slab at its top surface, horizontal and vertical supports are applied with the use of rigid plates of dimensions (50 mm x 50 mm x 10 mm) and frictionless contact, at a distance of 3c from the symmetry plane (see also left part of Figure 1).

2 Example finite element discretization

The model is constructed and discretized with the help of Cubit [1] and then exported to Abaqus [2] format.

20 node quadratic brick finite elements with reduced integration for the concrete (C3D20R), linear 8 noded brick finite elements for the steel plate and anchor and linear 8 noded cohesive elements (COH3D8) for the mortar are used.

The connection between the steel plate and the concrete is modeled by a hard contact interface law, i.e interpenetration is not allowed and the friction between the two bodies is not taken into account. The lateral connection between 1) anchor and mortar and 2) mortar and concrete is performed using tie constraints. The base of the anchor at the embedment height is free.

Mortar is modeled with the use of an elastic traction-separation bond law available in Abaqus, see Table 4. Table contains, k_nm mortar stiffness ([MPa/mm]) in normal direction, k_tm mortar stiffness ([MPa/mm]) in the tangential direction (same for both tangential directions), and mortar density ρ_m.

Parameter	Value	Units
k_nm	5400	MPa/mm
k_tm	2700	MPa/mm
ρ_m	2×10^{-6}	kg/mm^3

Table 4: Mortar properties

2.1 Solution procedure

An implicit dynamic analysis is performed using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation.

References

[1] Coreform LLC. Coreform cubit, 2025. Computer software.

[2] Dassault Systèmes SIMULIA. *Abaqus 2024 Documentation.* Dassault Systèmes, 2024. Finite element analysis software.

[3] Krešimir Ninčević and Roman Wan-Wendner. On the dependence of concrete edge breakout on concrete age and coarse aggregate type. *Structural Concrete*, 22(5):2952–2966, 2021.